

UJUNG PANDANG HASANUDDIN, Indonesia

WMO: 971800

Lat: 5.0711S

Lon: 119.5522E

Elev: 46

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 68.2	(c) 69.8	(d) 55.8	(e) 66.4	(f) 82.5	(g) 59.0	(h) 74.8	(i) 81.6	(j) 17.3	(k) 82.4	(l) 14.9	(m) 82.8	(n) 2.7	(o) 180	(p) 0.310

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 10	(b) 18.9	(c) 94.2	(d) 75.0	(e) 93.2	(f) 75.8	(g) 91.8	(h) 76.6	(i) 82.7	(j) 87.3	(k) 81.8	(l) 87.0	(m) 81.0	(n) 86.7	(o) 9.5	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.0	166.6	84.5	80.7	159.8	83.9	79.7	154.6	83.4	46.8	87.6	45.7	87.7	44.8	87.2	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.5	(b) 12.8	(c) 11.3	(e) 64.9	(f) 97.4	(g) 2.0	(h) 2.3	(i) 63.5	(j) 99.1	(k) 62.3	(l) 100.4	(m) 61.2	(n) 101.7	(o) 59.8	(p) 103.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			60.5	83.8	2.6	1.5	58.6	84.9	57.0	85.8	55.5	86.7	53.6	87.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	80.4	80.6	81.2	81.9	82.3	81.5	80.6	81.1	82.0	82.7	82.3	80.9
	DBStd	1.87	1.97	1.85	1.70	1.55	1.50	1.61	1.68	1.81	1.71	1.51	1.64	2.05
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11483	942	856	967	956	1001	945	949	965	960	1013	970	959
	CDD65	6008	477	436	502	506	536	495	484	500	510	548	520	494
	CDH74	60229	4093	3813	4661	4929	5542	4882	4777	5514	6094	6271	5262	4391
	CDH80	24166	1195	1191	1625	1835	2192	1917	2012	2590	3043	3027	2149	1390
(14) Wind	WSAvg	4.3	4.5	4.4	4.2	3.9	3.7	3.6	4.0	4.7	5.0	4.8	4.3	4.4
(15) Precipitation	PrecAvg	136.2	30.2	23.1	17.0	10.8	6.5	3.9	2.4	0.9	1.6	4.6	10.9	24.4
	PrecMax	204.5	51.2	46.5	29.8	22.2	13.1	8.7	12.2	4.5	13.7	16.3	27.0	41.7
	PrecMin	101.6	6.1	3.9	8.2	4.7	2.3	0.8	0.2	0.3	0.5	0.0	2.8	9.1
	PrecStd	27.7	9.2	9.5	6.5	4.0	2.8	2.5	2.8	1.0	2.5	4.1	5.7	8.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.0	90.4	91.6	92.3	92.8	92.9	92.7	94.5	95.8	96.9	94.0	91.6
		MCWB	81.0	79.7	80.4	80.1	79.5	77.7	76.2	74.8	72.7	72.7	78.1	80.2
	2%	DB	89.1	89.3	89.9	91.1	91.5	91.2	91.3	92.9	93.9	94.3	92.2	89.8
		MCWB	79.7	79.6	79.8	79.9	79.3	77.6	76.1	74.4	74.1	74.9	78.4	79.9
	5%	DB	87.7	87.9	89.2	89.7	90.0	89.7	89.8	91.5	92.9	93.0	91.2	88.3
		MCWB	79.4	79.4	79.5	79.3	78.9	77.8	76.0	74.7	74.4	75.7	78.6	79.6
	10%	DB	86.1	86.3	87.7	88.2	89.3	88.2	88.4	89.9	91.5	91.5	89.7	87.3
		MCWB	78.9	79.0	79.1	79.0	78.8	77.5	75.8	74.4	74.4	76.4	78.8	79.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.1	82.7	83.0	83.1	82.9	81.8	80.3	79.5	81.0	82.1	83.5	82.9
		MCDB	87.2	86.2	87.5	87.3	87.8	86.6	87.1	87.6	87.0	87.4	88.4	87.1
	2%	WB	81.2	81.2	81.4	81.6	81.7	80.7	79.2	78.3	79.2	80.4	81.8	81.6
		MCDB	85.9	85.9	86.8	87.4	87.3	86.5	85.9	86.7	86.7	87.5	87.4	86.5
	5%	WB	80.2	80.2	80.5	80.7	80.8	79.8	78.4	77.4	78.1	79.5	80.6	80.6
		MCDB	85.0	85.3	86.2	86.9	86.8	85.5	85.2	86.4	86.3	86.9	87.1	86.2
	10%	WB	79.3	79.4	79.7	79.9	79.9	79.0	77.6	76.4	77.0	78.7	79.7	79.7
		MCDB	83.7	84.2	85.0	85.7	85.7	84.6	84.5	85.8	85.9	86.3	86.1	84.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.9	13.0	14.0	14.9	15.5	17.5	19.7	20.7	18.9	15.2	11.9
		MCDBR	13.5	14.0	15.0	15.4	16.1	17.3	19.3	21.3	22.7	21.8	17.6	14.3
		MCWBR	6.1	6.4	6.7	6.8	6.9	7.0	7.8	8.6	9.5	9.3	7.9	6.5
		MCDBR	12.2	13.0	13.6	14.4	14.8	15.1	16.4	18.6	18.8	17.3	15.7	13.3
	5% WB	MCWBR	6.1	6.4	6.5	6.9	6.8	6.7	7.2	8.2	8.5	7.7	7.5	6.7
(40) Clear-Sky Solar Irradiance	taub		0.435	0.435	0.427	0.418	0.416	0.417	0.407	0.416	0.436	0.477	0.476	0.442
	taud		2.422	2.404	2.434	2.473	2.481	2.481	2.500	2.445	2.366	2.242	2.270	2.402
	Ebn at Noon		287	286	285	280	272	267	272	276	278	271	273	284
	Edh at Noon		39	40	38	36	34	33	33	36	40	46	45	40
(44) All-Sky Solar Radiation	RadAvg		1461	1547	1684	1710	1648	1537	1660	1910	2085	2059	1822	1475
	RadStd		140	131	110	101	80	124	117	97	104	165	140	134

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.43	N/A	N/A	+0.99	+1.30	+1.58	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.43	N/A	N/A	+0.99	+1.30	+1.58	N/A	N/A	N/A	N/A

Nomenclature: See separate page